De Mathematics Competitions

4th Annual

DMC 12 S

Sunday, September 27, 2026



INSTRUCTIONS

1. DO NOT TURN TO THE NEXT PAGE UNTIL YOUR COMPETITION MANAGER TELLS YOU TO BEGIN.
2. This is a 25-question multiple-choice competition. For each question, only one answer choice is correct.
3. Mark your answer to each problem on the answer sheet with a #2 pencil. Check blackened answers for accuracy and erase errors completely. Only answers that are properly marked on the answer sheet will be scored.
4. SCORING: You will receive 6 points for each correct answer, 1.5 points for each problem left unanswered, and 0 points for each incorrect answer.
5. Only blank scratch paper, rulers, compasses, and erasers are allowed as aids. No calculators, smartwatches, phones, or computing devices are allowed. No problems on the competition will require the use of a calculator.
6. Figures are not necessarily drawn to scale.
7. Before beginning the competition, your competition manager will ask you to record your name on the answer sheet.
8. You will have 75 minutes to complete the competition once your competition manager tells you to begin.
9. When you finish with the competition, sign your name in the space provided on the answer sheet and complete the demographic information questions on the back of the answer sheet.

The problems and solutions for this DMC 12 were prepared
by the DMC Committee under the direction of:

Taiki Aiba and Ryan Shin

The DMC Committee reserves the right to disqualify scores from a school if it determines that the rules or the required security procedures were not followed.

The publication, reproduction or communication of the problems or solutions of this competition during the period when students are eligible to participate seriously jeopardizes the integrity of the results. Dissemination via phone, email, or digital media of any type during this period is a violation of the competition rules.

Students who score well on this DMC 12 will be invited to take the De Invitational Mathematics Examination (DIME), which is already out and is due on Friday, January 29, 2027. More details about the DIME can be found on our site.

1. Let $f(x) = \log_9 x$, and let $g(x) = f(x^2)$ for all nonzero real numbers x. Which of the following is equal to $g(x)$ for all nonzero real x?

   (A) $\log_3 x$ **(B)** $\log_3 |x|$ **(C)** $2\log_3 |x|$ **(D)** $\log_9 \sqrt{x}$ **(E)** x^2

2. What is the least positive integer n such that $n! + 1$ is not divisible by any integer from 2 to 9, inclusive?

   (A) 4 **(B)** 5 **(C)** 6 **(D)** 7 **(E)** 8

3. A rectangle with perimeter 40 is divided into four smaller rectangles by one line parallel to each pair of its sides. Three of the smaller rectangles have perimeters 19, 21, and 25. What is the perimeter of the fourth?

   (A) 9 **(B)** 15 **(C)** 17 **(D)** 23 **(E)** 27

4. Real numbers x and y satisfy $\sqrt{x\sqrt[3]{y}} = 4$ and $\sqrt{y\sqrt[3]{x}} = 9$. What is xy?

   (A) 36 **(B)** 72 **(C)** 108 **(D)** 144 **(E)** 216

5. Daniel walks one mile in four straight segments, each a quarter mile long. After each of the first three segments, he turns 90° either clockwise or counterclockwise, each with probability $\frac{1}{2}$. What is the probability that he ends where he started?

   (A) $\frac{1}{8}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{2}$ **(D)** $\frac{5}{8}$ **(E)** $\frac{3}{4}$

6. The numbers a, b, and c are three consecutive positive integers in some order, and so are p, q, and r. Also, $a \cdot p = 161$ and $b \cdot q = 189$. What is $c \cdot r$?

   (A) 128 **(B)** 144 **(C)** 160 **(D)** 176 **(E)** 192

7. For which values of k does the equation

   $$k^2x + 2 = 4x + k$$

   have no real solutions x?

   (A) -2 **(B)** 0 **(C)** 2 **(D)** -2 and 0 **(E)** -2 and 2

8. Joe writes a positive integer on a piece of paper. Each minute, he erases the number on the paper, adds 1, 2, or 3 to it, doubles the result, and writes down the new number. After four minutes, the number on the paper is 780. What is the sum of the digits of the least integer Joe could have started with?

   (A) 6 **(B)** 7 **(C)** 8 **(D)** 9 **(E)** 10

9. Kevin chooses two different positive integers and writes their sum, their positive difference, and their product on a board. Exactly one of the three numbers on the board is even, and it is 72. What is the sum of all possible values of the smaller of Kevin's two integers?

   (A) 5 **(B)** 6 **(C)** 8 **(D)** 11 **(E)** 12

10. In rectangle $ABCD$ with $AB = 5$ and $BC = 4$, point E lies on side $\overline{CD}$ so that segment $\overline{AE}$ bisects $\angle BED$. What is AE?

    (A) $3\sqrt{2}$ **(B)** $2\sqrt{5}$ **(C)** $3\sqrt{3}$ **(D)** $2\sqrt{7}$ **(E)** $\sqrt{30}$

11. Real numbers a, b, and c satisfy

$$\begin{aligned} a + b + c &= 1, \\ a + |b| + |c| &= 4, \\ |a| + b + |c| &= 5, \\ |a| + |b| + c &= 8. \end{aligned}$$

    What is $a^2 + b^2 + c^2$?

    (A) $\frac{53}{2}$ **(B)** $\frac{55}{2}$ **(C)** $\frac{57}{2}$ **(D)** $\frac{59}{2}$ **(E)** $\frac{61}{2}$

12. Aki and Hanami each have a positive integer from 1 to 20, inclusive. Each knows only their own number and that at least one of the two numbers is odd. Aki says, "My number is greater than 4 but less than 7." Hanami says, "Now I know that my number is at least three times yours." Aki says, "Now I know your number." What is the sum of Aki's and Hanami's numbers?

    (A) 21 **(B)** 22 **(C)** 23 **(D)** 24 **(E)** 25

13. In a classroom, each student has exactly two pencils, and each pencil is orange, yellow, or gray. Exactly 56 students have at least one orange pencil, 48 students have no gray pencil, and 34 students have two pencils of different colors. How many students have at least one yellow pencil?

    (A) 26 **(B)** 27 **(C)** 28 **(D)** 29 **(E)** 30

14. A plane intersects a sphere of radius 11 in a circle of area 108π. A second plane, perpendicular to the first, intersects the sphere in a circle of area 94π. The two planes meet in a line. What is the length of the part of this line inside the sphere?

    (A) $6\sqrt{3}$ **(B)** 12 **(C)** $11\sqrt{2}$ **(D)** $8\sqrt{5}$ **(E)** 18

15. Define the polynomial

$$P(x) = (x-a)(x-b)(x-c)(x-d),$$

where a, b, c, and d are positive real numbers less than 1. What is the greatest possible number of real numbers r such that $|P(r)| = 1$?

(A) 0 **(B)** 2 **(C)** 4 **(D)** 6 **(E)** 8

16. Bill and Ben each start with 2 fair coins. On each turn, both of them flip all the coins they have at the same time. Each coin that lands heads is given to the other person, and each coin that lands tails stays with its owner. What is the probability that Bill and Ben each have 2 coins after the third turn?

(A) $\frac{1}{4}$ **(B)** $\frac{5}{16}$ **(C)** $\frac{3}{8}$ **(D)** $\frac{7}{16}$ **(E)** $\frac{1}{2}$

17. Let $w = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $z = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$. How many ordered pairs (a, b) of positive integers with $a \leq 6$ and $b \leq 6$ satisfy $|3w^a + 4z^b| = 5$?

(A) 4 **(B)** 6 **(C)** 8 **(D)** 12 **(E)** 16

18. Parallelogram $ABCD$ has $BC = 5$, $\angle ABC < 90°$, and $\angle ACB > 90°$. The circle through A, B, and C meets line AD again at P and side $\overline{CD}$ again at Q. Also, $CP = 10$ and $CQ = 4$. What is AP?

(A) $\frac{48}{7}$ **(B)** 7 **(C)** $\frac{36}{5}$ **(D)** $\frac{15}{2}$ **(E)** 8

19. In equilateral $\triangle ABC$, points D and E lie on lines AB and AC, respectively, both on the opposite side of line BC from A. Also, $CE = DE$, and the circumcircle of $\triangle CDE$ is tangent to line AB at D. What is the degree measure of $\angle CDE$?

(A) 70 **(B)** 72 **(C)** 75 **(D)** 80 **(E)** 84

20. Each term of a sequence $a_1, a_2, \ldots, a_{2026}$ is -2, -1, 0, 1, or 2, and

$$\begin{aligned} a_1 + a_2 + \cdots + a_{2026} &= 20, \\ a_1^2 + a_2^2 + \cdots + a_{2026}^2 &= 100, \\ a_1^3 + a_2^3 + \cdots + a_{2026}^3 &= 50. \end{aligned}$$

What is the least possible value of $a_1^4 + a_2^4 + \cdots + a_{2026}^4$?

(A) 150 **(B)** 160 **(C)** 170 **(D)** 180 **(E)** 190

21. Isosceles trapezoid $ABCD$ with $\overline{AB} \parallel \overline{CD}$, $AB = AD = BC = 3$, and $CD = 7$ is inscribed in a circle. Point E on the circle satisfies $\overline{BE} \perp \overline{AC}$. What is DE^2?

(A) 24 **(B)** 26 **(C)** 28 **(D)** 30 **(E)** 32

22. What is the sum of the digits of the least positive integer n such that

$$\sqrt{5n-1} - \sqrt{5n-2} + \sqrt{5n-3} - \sqrt{5n-4}$$

is less than 0.05?

(A) 8 **(B)** 9 **(C)** 10 **(D)** 11 **(E)** 12

23. In right $\triangle ABC$ with a right angle at A and $BC = 7$, let D be the midpoint of side $\overline{AB}$. There is a point E on the circumcircle of $\triangle ABC$ such that $\triangle CDE$ is equilateral. What is the side length of $\triangle CDE$?

(A) $\sqrt{21}$ **(B)** $2\sqrt{7}$ **(C)** $\frac{7\sqrt{21}}{6}$ **(D)** $\frac{7\sqrt{10}}{4}$ **(E)** $3\sqrt{6}$

24. How many ordered pairs of positive integers (m, n) satisfy the following?

"There are exactly m sets of 100 consecutive positive integers whose least element is less than 100 that contain exactly $m + 1$ multiples of n."

(A) 3 **(B)** 4 **(C)** 5 **(D)** 6 **(E)** 7

25. A sequence $a_1, a_2, \ldots$ satisfies $a_1 = 2$, $a_2 = 9$, and $a_{n+2} = 2a_{n+1} + a_n$ for all positive integers n. The sum

$$\sum_{k=1}^{\infty} \frac{a_k a_{k+1}}{(a_{k+1}^2 - a_k^2)^2}$$

equals $\frac{1}{m}$ for a positive integer m. What is the sum of the digits of m?

(A) 16 **(B)** 17 **(C)** 18 **(D)** 19 **(E)** 20